Market Regime Detection and Dynamic Portfolio Allocation

A regime-based systematic trading strategy using a Hidden Markov Model to identify latent market states and dynamically size positions in SPY based on the estimated probability of a risk-on regime.

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Data Range	January 2005 - Present
Model	Gaussian HMM (K = 2, full covariance)
Universe	SPY, TLT, GLD, VIX
Benchmark	SPY Buy-and-Hold

1. Overview

Financial markets do not evolve as a single stationary process. They periodically shift between distinct regimes: calm, trending periods characterised by low volatility and positive momentum, and stress periods marked by elevated volatility, large drawdowns, and flight-to-safety flows. Identifying the current regime in real time enables a portfolio manager to modulate risk exposure rather than maintaining a fixed allocation throughout the cycle.

This project models those regimes in an unsupervised fashion using a Gaussian Hidden Markov Model (HMM) trained on a multi-asset feature set. Rather than assigning hard regime labels, the model outputs a continuous probability of being in a risk-on state. That probability is used to scale SPY exposure dynamically, with leverage capped at 2x and 5 basis-point-per-unit transaction costs applied to every rebalance.

2. Project Structure

```
MarketRegimeDetection/
    data/
        raw/market_prices.csv
        processed/features.csv
    notebooks/
        01_data_collection.ipynb
        02_feature_engineering.ipynb
        03_hmm_modelling.ipynb
    requirements.txt
    README.md
```

The pipeline is split across three sequential notebooks. Each notebook writes its output to disk so the next step can be run independently.

3. Data

Daily adjusted close prices are downloaded via the **yfinance** library starting 2005-01-01. Four instruments are collected to capture both equity risk and cross-asset regime signals:

Ticker	Asset	Role in feature set
SPY	SPDR S&P 500 ETF	Primary trading vehicle and core feature source
TLT	iShares 20+ Year Treasury ETF	Flight-to-safety / risk-off signal
GLD	SPDR Gold ETF	Safe-haven / inflation signal
^VIX	CBOE Volatility Index	Equity fear gauge; used as a level feature

A single missing observation per equity ticker (one non-trading day) is filled forward. The raw prices are saved to *data/raw/market_prices.csv*.

4. Feature Engineering

Nine features are constructed to capture the multi-dimensional nature of market regimes. Features span volatility, trend, drawdown, cross-asset momentum, and the fear gauge:

Feature	Description
SPY_vol_20	20-day annualized rolling standard deviation of SPY daily returns
SPY_price_ma50	SPY price divided by its 50-day simple moving average
SPY_price_ma200	SPY price divided by its 200-day simple moving average
SPY_drawdown	Drawdown of SPY from its rolling all-time high
VIX_level	Raw daily close of the VIX index
TLT_ret_20	20-day return of TLT (captures bond market flight-to-safety)
GLD_ret_20	20-day return of GLD (captures safe-haven demand)
SPY_ret_5	5-day return of SPY (short-term momentum)
SPY_ret_20	20-day return of SPY (medium-term momentum)

The 200-day moving average window means the feature matrix begins 2005-10-17. All NaN rows are dropped before modelling. The resulting dataset spans roughly 5,100 trading days. Features are standardised using StandardScaler fitted on the training split and applied to both splits.

5. Methodology

5.1 Train / Test Split

The feature matrix is sorted chronologically and split 70 / 30. No shuffling is applied; the split respects temporal order to prevent look-ahead bias. The StandardScaler is fit exclusively on the training set and then applied to both sets.

5.2 Choosing the Number of Regimes

Three complementary diagnostics were used to select K:

Elbow method on KMeans inertia: the elbow in the inertia curve suggested K = 4 as a plausible candidate.

Silhouette score: averaged over all samples, K = 2 produced the highest silhouette score, indicating well-separated, compact clusters.

PCA cluster visualisation: projecting the training set onto its first two principal components showed K = 3 and K = 4 to produce heavily overlapping clusters. K = 2 gave the clearest separation.

K = 2 was selected.

5.3 Gaussian HMM

A Gaussian HMM with full covariance matrices is trained on the standardised training data. Full covariance allows the model to capture correlations between features within each regime state.

```
GaussianHMM(n_components=2, covariance_type='full', n_iter=1000,
random_state=42).fit(X_train)
```

After fitting, **predict_proba** is called on the full (scaled) dataset to obtain a daily time series of regime probabilities. These soft probabilities drive position sizing rather than discrete regime labels.

6. Regime Interpretation

The two regimes were characterised by computing mean feature values grouped by regime label on the training set:

Regime	SPY 20d Return	SPY Volatility	VIX Level	SPY Drawdown	Interpretation
0	-0.25%	22.6%	25.6	-17.7%	Crisis / Risk-off
1	+1.40%	10.3%	14.0	-1.2%	Bull / Risk-on

Regime 0 corresponds to market stress: negative returns, volatility above 20%, VIX near 25, and deep drawdowns. Regime 1 captures the normal bull market environment with a VIX near 14 and minimal drawdown.

7. Historical Regime Validation

To validate that the model correctly identifies known crisis periods, the share of days classified as Regime 0 (crisis) was computed for three distinct historical episodes:

Episode	Window	Regime 0 (Crisis) Share
2008 Global Financial Crisis	Jan 2008 - Jun 2009	100%
COVID-19 Crash	Feb 2020 - Jun 2020	83%
2022 Inflation Shock	Jan 2022 - Oct 2022	94%

All three episodes were overwhelmingly classified as Regime 0, confirming that the HMM learns economically meaningful states rather than arbitrary statistical clusters.

8. Strategy Logic

Position size is a continuous, monotone function of the risk-on probability. This avoids the whipsawing that comes from hard regime switches and scales exposure proportionally to conviction:

```
position = clip(p_risk_on * 2, 0, 2)
```

At p_risk_on = 1.0 the strategy holds 2x leveraged SPY. At p_risk_on = 0.5 it holds 1x. At p_risk_on = 0.0 it is flat. The position is executed on the next day's open (shifted by one period).

Transaction Costs

A per-unit transaction cost of 5 basis points is charged on every change in position size:

```
turnover = |position - position.shift(1)|
transaction_cost = 0.0005 * turnover
strategy_return = position.shift(1) * SPY_ret - transaction_cost
```

9. Backtest Results

Performance is evaluated over the full dataset from late 2005 to the present, covering the 2008 financial crisis, the 2020 pandemic crash, the 2022 rate-hike cycle, and the 2025-26 tariff shock:

Metric	Buy and Hold (SPY)	Regime Strategy
Sharpe Ratio	0.65	0.79
Sortino Ratio	0.61	0.80
Maximum Drawdown	-55%	-30%
Total Return	8.04x	9.99x

The regime strategy improves the Sharpe ratio from 0.65 to 0.79 and the Sortino ratio from 0.61 to 0.80, while cutting maximum drawdown nearly in half (from -55% to -30%). Total return over the full period increases from 8.04x to 9.99x despite the drag from transaction costs and periods of reduced or zero exposure.

The reduction in maximum drawdown is the most practically significant result. A 30% peak-to-trough loss is materially easier to sustain and recover from than a 55% loss, both psychologically and in terms of the return required to get back to the prior high (43% vs 122%).

10. Installation and Usage

Clone the repository and install dependencies:

```
git clone https://github.com/naamShahreyar/MarketRegimeDetection.git
cd MarketRegimeDetection
pip install -r requirements.txt
```

Run the notebooks in order:

```
1. notebooks/01_data_collection.ipynb
2. notebooks/02_feature_engineering.ipynb
3. notebooks/03_hmm_modelling.ipynb
```

Each notebook writes its output to the *data/* directory so that subsequent notebooks can be run independently without re-executing earlier steps.

11. Dependencies

Package	Purpose
pandas	Data manipulation and time-series alignment
numpy	Numerical operations and array math
yfinance	Downloading historical market data from Yahoo Finance
matplotlib	Visualisation of price series, regimes, and backtest curves
seaborn	Statistical plots (boxplots, heatmaps)
scikit-learn	StandardScaler, KMeans, PCA, silhouette_score
hmmlearn	Gaussian HMM implementation